

Chapter 8

The Effects of Materials Technology and Fabrication Tolerances on Guided-Wave Optical Communication and Signal Processing

WILLIAM S. C. CHANG

Department of Electrical Engineering and Computer Sciences
University of California, San Diego
La Jolla, California

I. Introduction	348
II. Optical Guided-Wave Components in Communications and Signal-Processing Systems	349
III. The Effect of Fabrication Techniques on the Performance of Optical Guided-Wave Components	351
A. The Optical Waveguide	352
B. Refractive Components in Planar Waveguides	355
C. Beam Control by Periodic or Quasi-Periodic Structures	358
D. The Directional Couplers and the Channel Waveguide Switches	362
E. Branched-Channel Waveguide Coupling and Switching	365
F. Optical Laser Sources and Detectors	367
IV. Conclusion	369
References	370

I. INTRODUCTION

With the advent of lasers, optical fibers, holography, optical information processing, guide-wave optics, high-speed photodetectors, and integrated optical-electronic devices, the use of optical systems for high-data-rate communication and signal processing has become very attractive. The five primary advantages of optical systems are (a) the large bandwidth (i.e., high speed), (b) isolation from electromagnetic disturbances, (c) low attenuation (i.e., long-distance communication without any repeater stations), (d) small size (light weight), and (e) the parallel-processing capabilities. The potential uses for optical communications and signal processing are very large. However, there are two major areas where a great deal of research must be done before the full potential can be realized.

The first research area is concerned with how to meet the stringent sub-micron tolerances required for the fabrication of guided-wave structures. For example, small radius of curvature, volume defects, or edge roughness of any optical circuit will cause radiation losses. Frequently, device structures with lines that are microns wide and millimeters long must be fabricated without a single defect in each line. Submicron variations of the fabricated device patterns from their theoretical design may limit the extinction ratio, the efficiency, the minimum insertion loss, or the signal-to-noise ratio of guided-wave devices. At the present time, the fabrication techniques used for guided-wave optics are primarily the ones adapted from the semiconductor industry. The tolerances of these techniques are often a significant fraction of the optical wavelength. These fabrication difficulties have led to high costs, cumbersome design, and nonideal performance.

The second research area is concerned with the improvement of the programmability of optical communication and signal-processing systems. Present optical systems are mostly systems designed for a single function, such as Fourier analysis. This nonprogrammability has, at the present time, limited the applications of optical and signal processors to special situations such as rf spectrum analysis.

What we need is an optical-electronic signal-processing system efficiently coupled together, which will take advantage of the benefits available in the optical system and will extend the flexibility of the optical systems by electronics. The key to the realization of a combined electronic-optical system is the ability to fabricate on the same chip electronic and optical devices and circuits that are effectively interconnected with each other. The realization of such a capability depends again on the progress that can be made in materials and processing technology.

In the following sections, we shall introduce the present role of guided-wave components in communications and signal processing, followed by a consideration of the various processing and materials technologies and their effects on the performance of guided-wave components.

II. OPTICAL GUIDED-WAVE COMPONENTS IN COMMUNICATIONS AND SIGNAL-PROCESSING SYSTEMS

In optical waveguides, the electromagnetic energy is confined within a small area of the waveguide, which is usually of the dimension of several micrometers or less. The confinement of the electromagnetic energy within such a small region close to the surface has allowed us to obtain efficient deflection, switching, and modulation with very little drive power. It has allowed us to interface waveguides effectively to fibers, laser sources, and photodetectors. It has also allowed us to fabricate various guided-wave structures by planar microfabrication techniques. Similar to microwave waveguides, optical waveguides are basically single-mode devices. Both the phase and the amplitude of the guided wave can be used to carry information. Optical waveguide components differ from the microwave components in that the control of wave propagation is obtained by open dielectric structures (metals are too lossy), in that the fabrication tolerance may dictate their design, and in that parallel-processing techniques are applicable.

The applications of guided-wave components can be divided, according to the manner in which optical energy is controlled, into four categories: (a) the use of the confinement energy in optical waveguides for high-speed and efficient switching, modulation, generation, detection, or deflection of optical signals; (b) the use of the coherent phase interference or phase information for sensing or measuring various effects such as rotation, acoustic wave diffraction, and strain, which in turn may be an indication of some other condition; (c) the use of spatial Fourier transform and parallel processing in optical waveguides to obtain high-speed Fourier analysis, convolutions, correlations, etc.; and (d) the use of phase-synchronized interaction of the confined radiation of a phased array of elements to obtain reflection, focusing, collimation, filtering, and other transformations.

The potential role of optical waveguide components in communications and signal processing can be most clearly demonstrated by a few examples. Figure 1 illustrates a typical example in which switching in a channel waveguide is used for high-data-rate single-mode fiber-optical

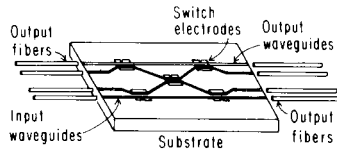


Fig. 1. Pictorial representation of 4×4 optical switching network.

networks [1,2]. By confining the optical energy to a few micrometers of transverse dimensions, large electric fields for high-speed switching can be achieved at a low applied voltage. Such a switch is one of the building blocks for a single-mode fiber-optical communications network [3]. Other components using different electrode and waveguide configurations have been demonstrated to obtain switching, modulation, or deflection of optical guided-wave signals for potential applications such as computer communications network [4], optical bistable circuits [5,6], and optical coding and decoding [7]. Figure 2 illustrates a fiber gyro in which the phase difference of the two counterpropagating waves are clearly distinguishable after a long propagation path. Such a gyro is very sensitive to rotation yet it occupies a small physical space [8,9]. Other physical sensors measuring pressures [10], strain [11], acoustic waves [12], electromagnetic field intensity [13], etc., have also been demonstrated by sensing the perturbation of the phase of the guided wave created by the environmental change. Figure 3 illustrates an rf spectrum analyzer. In this case, the confinement of both the optical and the acoustooptical energy near the surface has allowed us to achieve efficient acoustooptical diffraction with a moderate amount of acoustic power [14,15]. Parallel-processing capability of the analyzing lens and the detector array are used to obtain the spatial Fourier components of the diffracted beam. The spa-

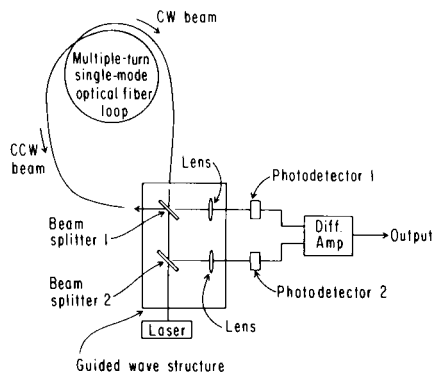


Fig. 2. Schematic illustration of a fiber gyro.

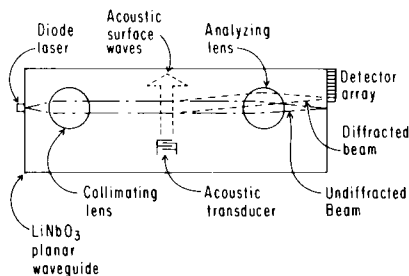


Fig. 3. Illustration of an rf spectrum analyzer.

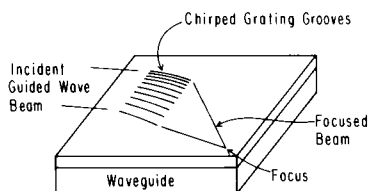


Fig. 4. Illustration of a chirped-grating lens.

tial Fourier content of the diffracted beam is proportional to the temporal Fourier content of the acoustic wave. Thus the signals from the detector array are a measure of the frequency spectra of the rf signal driving the acoustic transducer. Similar to the rf spectrum analyzer, convolution [16], pulse compression, correlation [17], and the decoding of various electric signals have also been demonstrated in optical waveguides utilizing the parallel-processing capability of optical systems. Figure 4 illustrates a chirped-grating lens that can be fabricated by planar processes. Here, phase-synchronization of the diffracted waves from each of the grating grooves is used to focus an incoming collimated guided wave to the focal point [18–20]. Similar gratings have been fabricated to provide feedback and frequency stabilization of GaAlAs lasers [21–24], wavelength demultiplexing and filtering in optical waveguides [25], input/output couplers [26,27], beam expanders [28,29], and so on.

III. THE EFFECT OF FABRICATION TECHNIQUES ON THE PERFORMANCE OF OPTICAL GUIDED-WAVE COMPONENTS

All optical guided-wave communication and signal-processing components and systems are composed of several basic elements:

- (1) A waveguide that confines the energy to a small region.
- (2) A refractive element that focuses, reflects, expands, or Fourier-analyzes a guided-wave beam by means of the variation of either the refraction index or the optical path (for example, a geodesic lens).
- (3) A phased array of dielectric discontinuities or variations that produce diffraction, reflection, cavity resonance, wavelength selection, beam focusing, etc. When such a phased array is controlled by electrooptical or acoustooptical effects, one obtains modulation, deflection, switching, etc.
- (4) A directional coupler consisting of two coupled, phase-synchronized, channel waveguides. By controlling the degree of phase synchronization, one controls the net power transfer between the two waveguides. When the phase-synchronization is controlled electrooptically, one obtains modulation or switching.
- (5) A channel waveguide branching coupler that either divides or combines the power into two channel waveguides according to a prescribed ratio.
- (6) A laser source or sources and associated electronics.
- (7) High-speed detectors or detector arrays and associated electronics.

In the following subsections, the effects of the fabrication tolerance and materials limitation on the performance of optical guided-wave components and systems will be assessed by analyzing the effects of fabrication methods on each of these elements and on their interconnection.

A. The Optical Waveguide

The simplest element is the optical waveguide. There are two kinds, the planar waveguide and the channel waveguide. In either case, a high-index layer ($\sim 1\ \mu\text{m}$ thick) or strip ($\sim 4\ \mu\text{m}$ wide and $1\ \mu\text{m}$ deep) is fabricated to confine the optical energy within its vicinity by means of a deposition, epitaxial growth, etching, diffusion, or ion-exchange process. Because of the optical, electronic, electrooptical, or acoustooptical properties required for switching, deflection, detection, or optical laser action, the commonly used waveguide materials are III-V compounds such as GaAs, GaAlAs, InGaAsP, or InGaAs, ferroelectrics such as LiNbO₃ or LiTaO₃, and high-index glasses on low-index glasses or SiO₂ substrates. The fabrication processes that are presently available to produce these waveguides are amorphous film deposition by evaporation, chemical vapor deposition (CVD), rf sputtering, wet and dry etching, diffusion and ion exchange, epitaxial growth, and ion milling.

The primary concerns regarding the properties of the fabricated waveguides are low attenuation and scattering. Waveguide materials are usually chosen so that they have little attenuation due to lattice vibration, carrier absorption, or impurity absorption. When the fabrication process introduces surface roughness or volume defects, scattering is created in the waveguide. Scattering causes attenuation of the waveguide mode (i.e., limits the throughput); it also converts the optical power from a desired mode to other unwanted modes (i.e., noise). For example, the conversion of the energy from a guided mode propagating in a given direction of the planar waveguide to other guided modes propagating in adjacent directions is the major source of in-plane scattering noise in signal processors [30]. Any rough surfaces created by wet chemical etching and sputter deposition of channel waveguides will create very large scattering losses.

1. Diffusion

Diffusion is the most commonly used fabrication process for materials such as LiNbO_3 because it smoothes out any rapid index variation along the waveguide boundary, thereby reducing the interface scattering. Single TE mode, planar LiNbO_3 waveguides have been obtained by Ti-indiffusion with less than 1-dB/cm attenuation and -25 -dB in-plane scattering noise at an angle of 4 mrad [31,32]. In-plane scattering noise for the TM_0 mode is about -35 dB at 4 mrad [31]. Channel waveguides have typically 1–2-dB/cm attenuation. So far, the performance of diffused waveguides has surpassed the performance of the epitaxially grown or deposited amorphous film waveguides on LiNbO_3 . However, diffused LiNbO_3 waveguides have several problems.

(1) When channel waveguides are fabricated by diffusion, there is lateral diffusion as well as vertical diffusion into the substrate.

(2) The Ti-indiffusion process is known to lower the optical damage level of LiNbO_3 , thereby limiting the power that can be transmitted. Frequently, optical damage can be observed for 1 mW of He–Ne beam (beamwidth 0.5 mm) propagating in such a LiNbO_3 waveguide that has 2- μm mode depth.

(3) For applications such as electrooptical switching with very small electrode spacing or the coupling of double heterostructure diode lasers to waveguides, it is desirable to have small mode depth (i.e., the optical energy is confined close to the waveguide surface). However, the minimum mode depth that can be realized in LiNbO_3 is limited to $\sim 2.5 \mu\text{m}$ (with $\Delta n \cong 0.015$ and diffusion depth = 1.5 μm), because long diffusion time will yield multimode waveguides, and within a limited diffusion time, only a small amount of Ti can be indiffused without leaving TiO_2 residue [33].

2. Ion Implantation

Ion implantation is capable of producing submicron resolution in the transverse dimension in semiconductor device fabrication. However, its application to the fabrication of optical waveguides has not yet been successful. This is partly because of the difficulty in implanting large atoms such as Ti to a depth of $1\text{ }\mu\text{m}$ into a material such as LiNbO_3 , the large flux density that is needed for changing the index, and the damage done by the implantation process. The only successful result reported is the creation of low-index area in LiNbO_3 by ion implantation [34,35].

3. Ion Exchange

Ion exchange has been used with mixed success to obtain waveguides in glasses and LiNbO_3 [36–39]. The attenuation of the Ag ion-exchanged single-mode glass waveguide was moderately high, caused by the large scattering. The attenuation of the Tl-exchanged multimode glass waveguide was very low. The attenuations of the Tl-exchanged single- and multiple-mode channel and planar LiNbO_3 waveguides have been excellent. Initial results obtained at the University of California, San Diego, for x -cut LiNbO_3 samples, single TE-mode planar waveguide and y -axis propagation are $n_{\text{eff}} - n_s = 0.018$, attenuation = 0.6 dB/cm , in-plane scattering noise of 35 dB at 3 mrad of angle, and no observable optical damage at 1 mW of He–Ne laser power with 0.5-mm beamwidth.

4. Evaporation, Chemical Vapor Deposition, and Sputtering

The deposition of amorphous films by e-beam evaporation, rf sputtering, and chemical vapor deposition has been very successful in providing both antireflection coating in optics and insulation layers in electronic devices. Thus it is a favored process for fabricating amorphous planar waveguides such as glass, Ta_2O_5 , Nb_2O_5 , and ZnO . However, the attenuation rate and the scattering loss that can be realized by this process vary a great deal from one material to another and depend critically on the environmental condition during deposition. Whenever the surface morphology is granular with surface roughness more than $100\text{ }\text{\AA}$, significant attenuation due to scattering may occur. The best results are obtained for glass planar waveguides deposited on a SiO_2 substrate (less than 0.6-dB/cm attenuation). The SiO_2 substrate is obtained from thermally oxidizing a Si wafer. However, channel waveguides fabricated by deposition of amorphous films combined with “lift-off” or with etching to obtain the channel pattern have not had low attenuation. The attenuation is caused primarily by the surface roughness of the side walls. There are two sources for this surface roughness, the roughness of the mask pattern and

the roughness created during etching. From this point of view, photolithography is not a desirable technique for making channel waveguide masks because it is difficult to make patterns with very large aspect ratio and it produces rough edges. Isotropic chemical etching is also known to produce rough wall surfaces. Research needs to be undertaken to investigate rf plasma or reactive ion-beam etching combined with e-beam lithography to produce low-loss single-mode channel waveguides.

5. Epitaxial Growth

Lattice-matched epitaxial growth by liquid-phase epitaxy (LPE), vapor-phase epitaxy (VPE), metal–organochemical vapor deposition (MOCVD), and molecular-beam epitaxy (MBE) are favored processes for making detectors, lasers, and electronic devices in III–V compound materials [40]. Although there is a lack of extensive data on waveguide attenuation fabricated by these processes, nevertheless, low-attenuation and low-scattering-loss planar waveguides can be expected. Data collected from preferential chemical etching and reactive ion etching of III–V semiconductor electronic devices also indicate that etching with smooth-wall and submicron resolution should be able to be obtained by these methods along certain crystallographic directions [41,42]. Masked epitaxial growth have also been known to create a mesa of GaAs with atomic smooth walls along specific crystalline directions [43]. Thus, we expect that low-loss channel waveguides could also be made by these processes. What needs to be developed is the materials growth and processing techniques that will allow waveguides and devices based on different alloy compositions to be grown on the same substrate and to be optically interconnected with each other with low loss.

B. Refractive Components in Planar Waveguides

For signal processing in planar waveguides, it is necessary to focus or to collimate a guided-wave beam by a lens. A lens is also used to obtain the spatial Fourier analysis of the guided wave, which provides the parallel-processing advantages in applications such as rf spectral analysis. In applications such as the acoustooptical convolver, the lens performs the function of spatial integration. In addition, a reflector is needed to fold a beam.

1. Refractive Waveguide Lenses

An ideal lens is an element that gives a quadratic phase shift $\exp[jk \times n_{\text{eff}} x^2 / 2f]$ to the incident beam. Here $k = 2\pi/\lambda$ is the propagation wave

number of a plane wave in free space, n_{eff} the effective index of the guided mode, f the focal length, and x the perpendicular distance away from the optical axis of the lens.

In bulk optics, a grinding process is used to produce the two curved surfaces of the lens. The quadratic phase shift is created by the refractive power of the lens material and the curvature of the lens surface. In waveguides, only the Luneberg lens (shown in Fig. 5a) is equivalent to a bulk lens [44]. The curved boundaries of the deposited layer serve the same functions as the curved surfaces of the bulk optical lens. Good performance has been obtained for the Luneberg lens on low-index waveguides by sputter-depositing a high-index layer through an aperture. However, the refractive power of any step discontinuity in a planar waveguide is given by the ratio of the n_{eff} on both sides of the step. Its value is very small for high-index waveguides such as LiNbO_3 because there are very few transparent materials that can be deposited on top with indices much larger than 2.2. Hence the Luneberg lens is not effective in high-index waveguides. The geodesic lens, as shown in Fig. 5b, is the most commonly used lens today [45,46]. In this lens, a geometrical indentation is created on the substrate. A waveguide is then fabricated to conform to the indented surface contour of the substrate. The differences in the geometrical path in different parts of the lens produce the desired quadratic phase shift. However, in order to reduce aberrations, nonspherical contours must be fabricated. It is also necessary to round off the corner of the

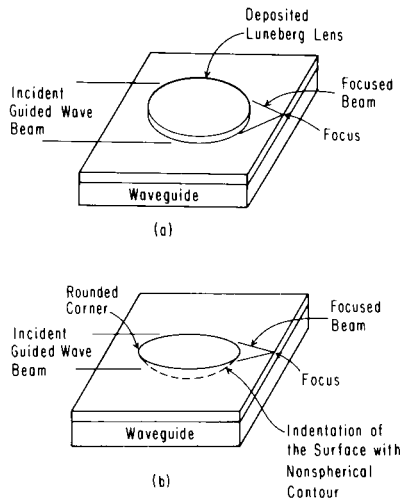


Fig. 5. Illustration of refractive lenses in optical waveguides: (a) the luneberg lens; (b) the geodesic lens.

rim of the circular indentation to a larger radius of curvature so that the radiation loss can be kept to a minimum. The major advantages of a geodesic lens are that (a) it has a wide field of view and (b) its performance does not vary rapidly with respect to wavelength.

In order to fabricate the indentation precisely, according to a calculated nonspherical contour, two methods have been used. The ultrasonic impact grinding employs a tool bearing the shape of the desired impression. Ultrasonic energy transmitted through the tool agitates the abrasive slurry and material is removed from the sample surface. The major task of the ultrasonic grinding process is to circumvent the tool wear that causes the fabricated surface contour to deviate from the design. Diamond turning employs a diamond-tipped tool to grind the surface depression. The grinding is controlled by precision slides and spindles, which are interfaced with a computer. The major tasks of diamond turning are the alignment of the tool, the determination of the location of the tool with respect to the center of the lens, and the determination of the tool shape that will give the theoretically designed contour. The waveguide is usually fabricated by one of the techniques described in Section III.A after the lens-grinding process. In this case, it is important to have a strongly guiding planar waveguide so that the radiation loss along the curved surfaces can be kept to a minimum. Excellent performances (diffraction-limited spot size with less than 2-dB insertion loss) have been obtained in diamond-turned lenses. The major disadvantages of geodesic lenses are that each lens must be ground separately (it is expensive) and that a deep indentation is required for large-diameter lenses.

2. Total Internal Reflectors

In order to create efficient reflection with a dielectric discontinuity, one must utilize total internal reflection. This means that there must be an edge that has an end surface perpendicular to the waveguide surface. The n_{eff} on the outside must be lower than the n_{eff} in the inside of the edge. If the edge has either micron-sized defects on its corners or a surface roughness of the order of a few hundred angstroms or more, there will be substantial random scattering loss. If the orientation of the end surface differs from the perpendicular direction by more than a few degrees, there will be energy reflected into the substrate modes. Thus techniques such as isotropic wet chemical etching and rf sputtering are unsuitable for the fabrication of that edge. So far, the only techniques that are successful in creating such a vertical end surface are (a) cleavage along a specific crystalline direction, (b) preferential chemical etching, (c) optical polishing, and (d) preferential epitaxial growth.

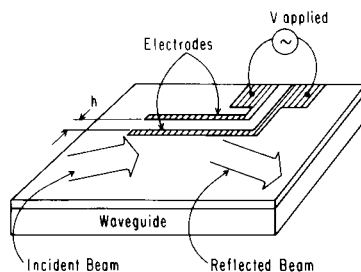


Fig. 6. The electrooptical total internal reflector.

3. Electrooptical Reflections

Reflection can also be obtained by the electrooptical effects in piezoelectric planar waveguides, as shown in Fig. 6 [47,48]. In this case, the orientation of the electric field produced by the applied voltage is such that it creates a reduction of the n_{eff} between the two electrodes. When the angle of incidence ψ' is larger than the critical angle, total internal reflection occurs. Since the critical angle is a function of the applied voltage, this is a reflector switch. This reflector differs from the reflectors made from a dielectric step in that it has very little random scattering loss produced by the edge roughness of the electrode pattern. No substrate mode will be excited. However, the electrooptical coefficient of the known materials at room temperature is limited to approximately 10^{-10} m/V. We need to minimize the spacing h between the electrodes in order to minimize the voltage required for efficient switching. The minimum h is limited by the mode depth of the waveguide. In some waveguides, more than 10 V of applied voltage may be necessary to obtain a high reflection coefficient [48]. The angle ψ' that can be realized is usually very small, depending on the amount of n_{eff} that can be reduced by the electrooptical effect.

C. Beam Control by Periodic or Quasi-Periodic Structures

Another well-known technique for obtaining deflection, focusing, reflection, etc., in bulk optics is diffraction. However, the diffraction efficiency of a single dielectric discontinuity in optical waveguides is very small. Thus the cumulative phase effect of a large number of dielectric discontinuities must be used to create the desirable diffraction efficiency. In order to obtain phase cumulation at a given wavelength, one must accurately position the dielectric discontinuities such that scattered guided waves from each discontinuity are in phase with each other along a spe-

cific direction of propagation. This is known as the phase-matching condition in Bragg diffraction. Many guided-wave components such as grating couplers [26,27], beam expanders [28,29], distributed feedback lasers [21–24], and grating lenses [18–20] are made from periodic or quasi-periodic structures. Here, the important fabrication consideration is the tolerance of various methods and the effect of tolerances on the performance of these periodic components.

1. The Bragg Reflector

The simplest periodic component is a Bragg grating reflector, shown in Fig. 7 [25]. In this case, we need to fabricate many identical and parallel grooves. These straight grooves have a constant periodicity Λ , where $\Lambda = m(\lambda/2n_{\text{eff}} \cos \theta)$, and m , called the *order*, is any integer greater than or equal to one and θ the angle of incidence of the guided-wave beam with respect to the plane normal to the grooves. Any guided-wave beam incident on the grating at the angle θ is reflected back with a power reflection efficiency η . In principle, the reflection can be obtained for all the Λ values corresponding to the various orders of m . In reality, it is desirable to have $m = 1$ because large Λ values that correspond to $m > 1$ will allow the guided-wave energy to be scattered into substrate and air radiation modes. Hence Λ is typically of the order of a fraction of a micrometer. Lithographic techniques with submicron linewidth must be used to make these gratings.

If $m = 1$ and if the random scattering due to the surface roughness of the grooves is neglected, $\eta = P_0 \tanh^2 \kappa L$, where L is the length of the grating reflector, κ the coupling coefficient, and P_0 the incident power [49]. The intensity of the transmitted beam is $P = P_0 \text{sech}^2 \kappa L$. If the λ is varied, then the reflectivity will vary because of the change in the phase-matching condition. The wavelength range $\Delta\lambda$ within which high reflection is obtained is inversely proportional to the grating reflector length L , provided $\kappa L \ll \pi$ [25]. Thus the design of a grating reflector is a determination of the κ , Λ , and L needed to achieve a prescribed reflec-

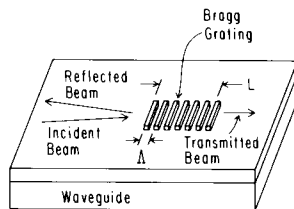


Fig. 7. The Bragg grating reflector.

tion and transmission over a given frequency range. If, in the process of fabrication, there is a chirping in the grating period $\Delta\Lambda$ or a variation in waveguide thickness corresponding to a change in the mode effective index Δn_{eff} , then the minimum $\Delta\lambda$ that can be realized is $\Delta\lambda/\lambda|_{\text{min}} \approx (\Delta n_{\text{eff}}/n_{\text{eff}}) + (\Delta\Lambda/\Lambda)$ [25]. For $\lambda = 5000 \text{ \AA}$ and $\Delta\lambda = 0.5 \text{ \AA}$, the fabrication requirements are $\Delta\Lambda/\Lambda \approx 10^{-4}$ and $\Delta n_{\text{eff}}/n_{\text{eff}} \approx 10^{-4}$. Note that the $\Delta\Lambda/\Lambda$ requirement implies a control of both the line position and the linewidth. Electron-beam lithography can only meet a $\Delta\Lambda/\Lambda$ requirement of the order of 10^{-2} . The only known method that can meet the stringent $\Delta\Lambda/\Lambda$ requirement is the exposure of the photoresist with the holographic interference pattern of two laser beams. In that case, $\Delta\Lambda/\Lambda = 10^{-5}$ has been obtained. The resist pattern from either the e-beam or the holographic laser process is transferred later to the waveguide by ion milling or by the "lift-off" process. In choosing the fabrication method for pattern transfer, care must also be taken to minimize both the distortion in linewidth and the random scattering loss due to the surface roughness of the grating grooves. The only known method that functions moderately well is ion milling. Radio-frequency plasma etching and reactive ion etching may provide better pattern transfer in the future. How to transfer the resist pattern is a serious problem that needs to be solved, especially for materials such as LiNbO_3 . Research is also needed to control the depth of the grating grooves according to a prescribed variation [28].

2. The Chirped-Grating Lens

Chirped gratings, as shown in Fig. 4, can also be used as a lens because different transverse sections of the grating will give a different angle of deflection so that a focused spot is created by the total lens [19,20]. The fabrication of such a chirped grating is even more difficult than that of a constant-periodicity grating reflector. For example, for a chirped-grating lens with $F = 10$, $\Lambda_{\text{max}} = 4 \text{ }\mu\text{m}$, $\Lambda_{\text{min}} = 2 \text{ }\mu\text{m}$, and 2-mm aperture size, the desired variation of the center positions of the adjacent grooves is on the order of tens of angstroms. The linewidth requirement of the fabrication technique is not high, but the positional accuracy requirement is very high.

In principle, chirped-grating patterns can be achieved holographically, but there is a great deal of practical difficulty in adjusting both the angles and the divergence of the two laser beams to obtain a specific chirping rate. The scanning electron-beam machine has sufficient resolution to do that, but the commercial e-beam lithographic machines have a quantization error of the order of $1000 \text{ \AA}$ for addressing their beam position. Their generated patterns are also limited to straight lines at right angles (any

curves must be approximated piecewise by many line segments at $1000\text{\AA}$ steps). The transfer of the resist pattern to the waveguide can be handled by the same techniques used to fabricate gratings with constant periodicity.

3. The Periodic Electrooptical Switch

Switching has also been demonstrated using periodic electrode structures on piezoelectric waveguides. For example, Bragg diffraction can be obtained electrooptically from the interdigital electrode shown in Fig. 8 [50]. When an rf voltage is applied to the electrode, its electric field creates a Bragg diffraction through the periodic index variation in the waveguide. The fabrication of electrooptically active waveguides such as LiNbO_3 has already been discussed in Section III.A. A buffer layer can be obtained by rf sputter deposition, evaporation, or chemical vapor deposition. Scattering loss and $\Delta\Lambda/\Lambda$ requirements are easy to satisfy for the fabrication of the electrode. Conventional photo- or e-beam lithography, followed by metallization, etching, or lift-off, can be used. However, we must consider three other effects of fabrication technology on the performance of an electrooptical Bragg switch.

(1) For small Λ , the mode depth of the available waveguide is a problem. If there is a substantial decrease of the electric field strength within the depth of the mode, there will be a significant reduction in the Bragg diffraction efficiency.

(2) Frequently, there are hundreds of fingers that are millimeters long and a few micrometers wide. For such a pattern, the yield of the photolithographic and metallization processes must be very high. A single short between any two fingers will render the entire device useless. From this

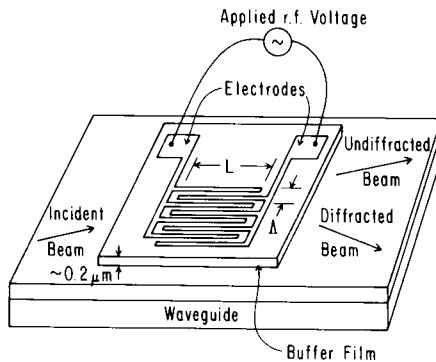


Fig. 8. The simple Bragg electrooptical deflector.

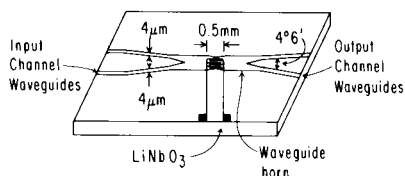


Fig. 9. Bragg electrooptical switch for channel waveguides.

point of view, e-beam and x-ray lithography are much superior to photolithography, not because of their resolution, but because they are much less sensitive to defects.

(3) In-plane scattering is a major source of noise for the Bragg diffraction [51].

Switching by Bragg electrode in piezoelectric channel waveguides, as shown in Fig. 9, has also been demonstrated [52]. Here a waveguide expansion horn is used to transfer a channel waveguide mode to a planar waveguide mode before Bragg switching takes place. A large extinction ratio (28 dB) has been obtained at 30 V of applied voltage in Ti-indiffused LiNbO_3 waveguides. There are three additional considerations concerning the fabrication of this structure.

(1) The expansion horns had an insertion loss caused by the radiation loss in the horn. The radiation loss is caused by the small difference $n_{\text{eff}} - n_s$ that can be realized in the LiNbO_3 waveguide and by the practical limit of the taper of the horn.

(2) Only a fraction of the diffracted beam is coupled into the second channel waveguide. The angle of intersection of the two channel waveguides is usually a few degrees to reduce the cross talk. Coupling into the second waveguide at such a large angle by the Bragg diffraction may be improved by means of small spacings and long electrode fingers. But the fabrication of small spacings or long electrode fingers will put more stringent requirements on lithography. For very small electrode spacings, the value of the overlap integral between the radio frequency and the optical electric fields will also be reduced because of the small $(n_{\text{eff}} - n_s)$.

(3) The angle of intersection of the two channel waveguides can be reduced without any increase in cross talk if the fabrication tolerance of the two crossing horns can be improved.

D. The Directional Couplers and the Channel Waveguide Switches

In principle, a directional coupler consists of two parallel channel waveguides with identical n_{eff} for their modes [53]. Figure 10 shows such a

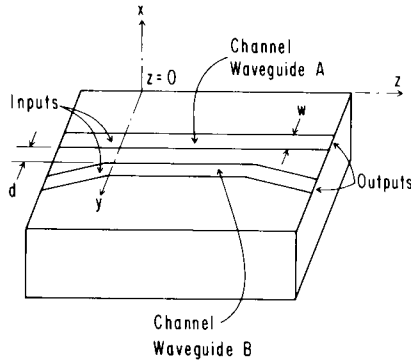


Fig. 10. The channel waveguide directional coupler.

device. The two waveguides are separated at a small distance d from each other such that d is less than the length of the evanescent tail D . The overlapping evanescent tails create a coupling between the two modes. When the power is entirely in the A mode at $z = 0$, the power conversion in A and B can be described as [49]:

$$P_B = I_0 \sin^2(\kappa z), \quad P_A = I_0 \cos^2(\kappa z),$$

where κ , the coupling coefficient, is proportional to $\exp(-d/D)$. If the coupling is terminated at a specific z value, a prescribed amount of power is transferred from A to B. When an electrode is fabricated on top of the waveguide in a piezoelectric material, as shown in Fig. 11, and an rf voltage is applied to the electrode, the electrooptical effect will create a difference in the n_{eff} of the modes in the two waveguides. The electrooptically controlled phase mismatch between the two modes will reduce the maximum power transfer from A into B. In this sense, any directional coupler in piezoelectric waveguides is also potentially an electrooptical switch [53].

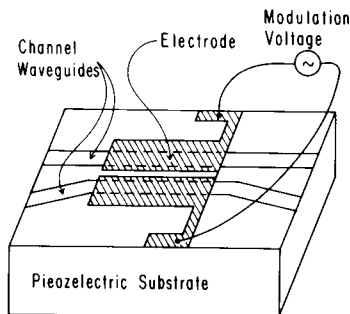


Fig. 11. The electrically switched optical directional coupler.

In order to fabricate a directional coupler, we need to fabricate channel waveguides and electrodes according to the patterns shown in Figs. 10 and 11. The mask patterns for directional couplers usually require an interaction length L of several millimeters, a d on the order of 2 or 3 μm and a guide width W on the order of 3 or 4 μm , depending on the substrate index and n_{eff} . It is difficult to fabricate such mask patterns by photolithography. Because of the small width and separation distance and the large aspect ratio of the length to the width, it is much easier to do it by e-beam lithography. The tolerance on waveguide sizes or indices that can be fabricated by deposition or etching using even an e-beam lithography mask is also very stringent. If there is a difference in n_{eff} corresponding to λ/L , there will be a substantial reduction of the power that can be transferred from A into B [53]. The coupling coefficient κ is also very sensitive to any variation in d because it is proportional to $\exp(-d/D)$. For Ti-indiffused LiNbO₃ waveguides, the lateral diffusion will create an additional uncertainty in κ . Thus, it is very difficult to fabricate directional couplers with a prescribed amount of power to be transferred. For the same reasons, electrooptical directional coupler switches made according to this design also have low extinction ratios. Better control in n_{eff} and in d can be obtained in planar waveguides. The most successful directional coupler is a planar waveguide deposited on top of another planar waveguide [54,55]. In that case, d and the waveguide thickness can be controlled by the deposition process to better than 100-Å accuracy. Good performances have been obtained. However, the planar waveguide directional coupler is not a useful device because it is very difficult both to interconnect such a structure with other waveguide components and to obtain electrooptical switching.

A number of design modifications have been made in order to circumvent the fabrication difficulties.

(1) A tuning electrode has been fabricated on the passive directional coupler in LiNbO₃ to fine tune the n_{eff} by means of the electrooptical effect [56,57].

(2) In the alternating electrooptical switch design, the electrode parameter (shown in Fig. 12) is split into two sections with alternating electric fields so that the effect of phase mismatch is reversed in the two sections. Cross-talk levels (i.e., extinction ratios) as low as -26 dB have been obtained at 20 V of applied voltage in this design [58]. In order to further reduce the applied voltage, multiple sections of alternating $\Delta\beta$ electrodes have been used [59]. The required voltage is 3.3 V for six sections of electrode. However, the extinction ratio has dropped to 13 dB because of the misalignment in the fabrication of the electrodes, and the rise time for the structure (with 22-pF capacitance, 0.017- μH inductance, and 12- Ω

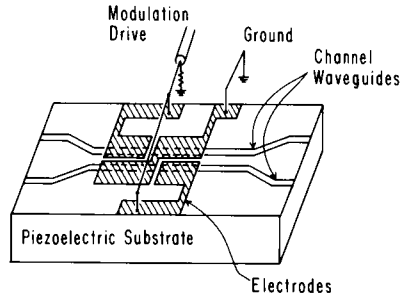


Fig. 12. The alternating $\Delta\beta$ electrooptical switch.

resistance) has risen to 2 nsec because of the inductive ringing associated with the lead.

(3) In order to compensate for the sensitivity to the polarization variation of the optical input into the channel waveguide, a second set of alternating electrodes that applies a horizontal electric field across the channel waveguides has been introduced [60].

Notice that a price has been paid for each compensation step.

E. Branched-Channel Waveguide Coupling and Switching

The channel waveguide branching coupler shown in Fig. 13 is used to divide the optical energy into separate channels or to recombine them into one channel according to a prescribed ratio [61]. Figure 14 shows an interferometer formed from two branched couplers. The energy transmitted

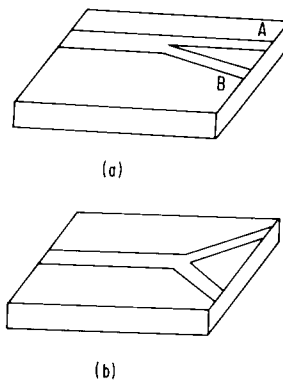


Fig. 13. The channel waveguide branching coupler: (a) asymmetric branch with small taper slope; (b) symmetric branch with large taper slope.

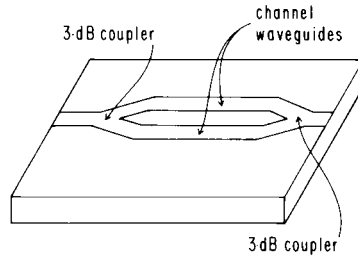


Fig. 14. The channel waveguide interferometer.

through such an interferometer depends on the phase difference of the signals in the two arms. The interferometer becomes a modulator when the phase difference is controlled by the electrooptical effects, and it is a sensor when the phase difference is produced by external conditions.

In principle, the abrupt branched coupler shown in Fig. 13b should be a power divider, e.g., dividing the optical energy into the two waveguides [61]. The asymmetrical coupler shown in Fig. 13a should be a power splitter; it should transmit the TE_0 -mode power into the waveguide A and TE_1 -mode power into the waveguide B [61]. When the electrodes are fabricated adjacent to the piezoelectric waveguides, the electrooptical change in index should convert a power divider into a mode splitter, thereby functioning as a modulator.

In reality, power division [62,63], mode splitting [62,63], y -channel modulation [64,65], branching waveguide modulation [66], interferometric modulation [67,68], etc., have all been experimentally demonstrated in channel waveguides, but none of them has a high extinction coefficient. The primary difficulty is the fabrication tolerance of the tapered interaction region (i.e., the control of the tapering angle, the index variation, the sharpness of the tapered point, and the shape of the taper). So far, the fabricated tapered sections are neither ideal power dividers nor ideal mode splitters. Unfortunately, the solution to these fabrication problems is very difficult to find. Even e-beam lithography only has a minimum resolution of 1000 Å. Attempts have been made to modify the power division in piezoelectric waveguides by the addition of an auxiliary tuning electrode on the side of the channel waveguides so that the index of the channel waveguide can be altered slightly by an applied voltage [63]. The auxiliary electrode did improve the extinction coefficient of the modulator, but the improved extinction coefficient is still worse than 20 dB. Perhaps the only possible approach is to combine a tapered velocity coupling with adiabatic modal evolution, as has been investigated in the planar waveguide modal-evolution 3-dB coupler [69]. In other words, we need both a redesign of the branching couplers to alleviate the fabrication requirements as well as an improvement in the control of fabrica-

tion tolerance in order to obtain a reasonable performance. The best performance of a channel waveguide interferometer switch was obtained by Leonberger [69a] with 17 dB extinction ratio, $V_\pi = 4.5$ V at $\lambda = 0.6328$ μm .

F. Optical Laser Sources and Detectors

Much research has been undertaken to investigate lasers, in particular, the semiconductor lasers, to be used as sources for optical-fiber communication [70]. Semiconductor diode lasers with oscillation wavelengths ranging from 0.8 to 1.5 μm , long lifetime, hundreds of megahertz of injection current modulation bandwidth, single-mode operation, and milliwatts of power output have been reported. Similarly, much research has been conducted in avalanche photodiodes, PIN photodiodes, and detector arrays for detection of optical radiation in the 0.8–1.5- μm wavelength range with picoseconds of response time and dark-current densities in the 10^{-5} – 10^{-8} -A/cm² range [71]. However, most of the research is concerned with optimizing the performance of the laser and the detector itself. Only a limited amount of effort has been expended so far to integrate the lasers or the detectors with optical waveguides [72–74]. The most common approach is to butt-couple a laser to a waveguide with a precision mechanical jig. Similar comments can be applied to detectors made from Si and III–V compound materials. It is well known that such a hybrid approach, e.g., GaAlAs lasers hybridly coupled to a LiNbO₃ waveguide or Si detector array butt-coupled to a LiNbO₃ waveguide, may allow us to optimize the performance of each component separately. On the other hand, the ability to fabricate the entire system (or subsystem) on the same substrate will substantially increase the reliability and the reproducibility of the system performance and may reduce the cost of manufacturing. Thus the integration of optical components on the same chip is still a highly desirable goal. The best-known material system in which such a “monolithic” guided-wave system may potentially be realized is the III–V compounds of GaAs, GaAlAs, GaInAs, GaInAsP, and InP grown by the various epitaxial growth methods. The primary difficulty in realizing such a “monolithic” approach is that different compositions of materials are required to be grown for different components [73,74]. For example, the alloy composition ideal for lasers or detectors will be useless for waveguides because of the optical absorption at that wavelength. Thus one must be able to selectively grow different materials in different spatial regions of that chip. In addition, the processes used to fabricate and to interconnect various components must have submicron resolution and must be compatible to each other. The sensitivity of the various guided-wave optical devices to the tolerance of the fabrication process has already been discussed in the previous subsections.

Recently a great deal of research interest has been generated to investigate electronic devices interconnected with lasers or detectors on the same chip so that electronic signal processing can be combined with optical sources and receivers [75,76]. The materials and fabrication considerations are similar to the considerations in interconnecting different optical devices together except for the following differences: In optoelectronic devices, we must consider the interface charge densities, the mobilities, the effect of heterostructures on the motion of charge carriers, etc., whereas in integrated optical devices, our main considerations are refractive index, absorption, random scattering loss, optical waveguide propagation mode profile, etc.

It is difficult to discuss in detail the effect of materials and fabrication technology on the performance of interconnected optical and electronic components since very little is known about them at this point. However, some general trends can be deciphered from the limited existing results.

(1) The most important optoelectronic material system is probably the lattice-matched materials of GaInAs and GaInAsP grown on InP. These materials have a band gap in the 1.3–1.5- μm region. Excellent epitaxial crystalline materials have already been grown by VPE, LPE, MBE, and MOCVD methods [77–80].

(2) From the experience gained in GaAs and GaAlAs research, we can draw the following conclusions:

(a) The most convenient method for fabricating an epitaxial layer with tapered thickness is the LPE method [74]. Tapered films are very effective in interconnecting one planar waveguide to another planar waveguide [81]. Tapered structures can probably also be grown by MBE through an aperture, but it will be very difficult to grow them by VPE and MOCVD.

(b) The method that hold the brightest promise for producing large-area uniform epitaxial layers with fine thickness control is the MOCVD method. The MBE method has been used successfully to fabricate extremely thin epitaxial layers with a large degree of freedom in alloy compositions.

(c) Masked epitaxial growth by MOCVD, VPE, and MBE can yield mesas that have submicron resolution and very fine edges. Vice versa, reactive ion etching and preferential chemical etching along certain crystalline directions may also be used to etch islands of epitaxial material with submicron resolution.

(d) The growth in the lateral direction can be utilized to create an epitaxial film that will have islands of insulator film sandwiched in between the epitaxial film and the substrate [82].

(3) The most likely electronic devices to be fabricated on ternary and quaternary III–V semiconductors are diodes and FETs. In the fabrication of these devices, the important processing techniques are ion implantation, growth of a low-interface charge-density insulation layer by CVD, lithographic and etching techniques that will yield submicron resolution, formation of the ohmic contacts, etc. Many of these techniques are discussed in other chapters of this book.

IV. CONCLUSION

The effects of materials technology and fabrication tolerance on guided-wave optical components and systems have been discussed in this chapter. These effects may be summarized as follows:

(1) Surface and interface roughness, volume defects, and absorption will lead to substantial insertion losses for propagation in optical waveguides.

(2) Limitations of the refractive indices of materials have forced us to use geometrical deformations for focusing and reflection that are very sensitive to surface contour deformations.

(3) Since the scattering effect from a single dielectric discontinuity is very small, a phased array of discontinuities must be used to achieve modulation, deflection, focusing, beam-splitting, filtering, etc. Especially stringent fabrication tolerance requirements in groove positions (much less than $0.1\ \mu\text{m}$) exist in the generation and in the pattern transfer of these periodic and quasi-periodic structures.

(4) Coupling, power division, and branching of channel waveguides present an especially stringent set of fabrication requirements for the uniformity of the guided-wave structures and the accuracy of the geometrical shapes (much less than $0.1\ \mu\text{m}$). The design of guided-wave structures that will alleviate these fabrication requirements is as important as improvements in processing techniques.

(5) Interconnection of different optical and electronic devices on the same chip requires extensive research in materials technologies to provide different alloy composition materials at different spatial locations and in processing technologies to be compatible with each other for the fabrication of very different devices.

A great deal of research is needed to develop new materials technology and fabrication processes and to minimize the fabrication tolerances. Vice versa, much more research must also be conducted on the design of

guided-wave components so that the effects of fabrication tolerance can be circumvented. The realization of these techniques and designs holds the key to the realization of the full potential of optical communications and signal processing.

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